NATURAL LANGUAGE PROCESSING

REPORT

N-GRAMS

ANIL CAN

SUMMARY

In this project, 1-gram, 2-gram and 3-gram analyses were performed on a dataset consisting of novel texts written in different languages (English and Turkish). The developed Java program reads the text file and processes the words in it, then calculates the n-gram frequencies over the word sequence and presents the most frequently occurring patterns to the user. In this way, the effects of language modeling and language differences at the n-gram level become observable.

The program is divided into three basic classes: FileProcessor, NGramGenerator and NGramAnalyzer. The FileProcessor class reads the .txt file in the given file path and obtains the entire content as a single text. Then, this text is converted to lowercase, purified from punctuation marks and divided according to the space character and converted into a word sequence. This preprocessing process allows for a consistent n-gram analysis regardless of the language.

The NGramGenerator class creates groups of the desired n size (1, 2 or 3 word combinations) using the received word sequence. In this class, the number of occurrences for each n-gram produced is counted and kept in a HashMap structure. This map is then sorted by frequency to find the first 10 most frequently occurring n-grams and printed to the console. In this way, the most characteristic patterns for each text can be easily observed.

All operations are controlled within the NGramAnalyzer class. This main class first reads and processes the file, then starts the 1-gram, 2-gram and 3-gram analyses respectively. The time elapsed for each analysis is calculated in milliseconds and performance measurements are made. These measurements are important in terms of examining the effects of the algorithm on the running time against different language structures.

As a result, the developed algorithm takes a text file from the user and performs n-gram analysis on this text and presents the analysis results to the user in the form of the most frequently occurring patterns and running time.

SCREENSHOTS

```java
package nlp1;

import java.util.Map;

public class Main {

	public static void main(String[] args) {
		String filePath = "grimms-fairy-tales_1.txt";  // test için buraya dosya yolu
		try {
			// 1. Dosyayı oku
			String text = FileProcessor.readFile(filePath);

			// 2. Ön işlem: kelimeleri çıkar
			String[] words = FileProcessor.preprocess(text);

			// 3. Her n (1, 2, 3) için işle
			for (int n = 1; n <= 3; n++) {
				System.out.println("\n=== " + n + "-gram Analysis === \n");
				long startTime = System.currentTimeMillis();

				Map<String, Integer> ngramCounts = NGramGenerator.generateNGrams(words, n);
				NGramGenerator.printTop10(ngramCounts);

				long endTime = System.currentTimeMillis();
				System.out.println("Running time: " + (endTime - startTime) + " ms\n");
			}

		} catch (Exception e) {
			System.out.println("Error: " + e.getMessage());
		}
	 }

}
```

Main Class

```java
package nlp1;

import java.util.*;

public class NGramGenerator {

    // n-gram üret ve frekansları döndür
    public static Map<String, Integer> generateNGrams(String[] words, int n) {
        Map<String, Integer> map = new HashMap<>();

        for (int i = 0; i <= words.length - n; i++) {
            StringBuilder sb = new StringBuilder();
            for (int j = 0; j < n; j++) {
                sb.append(words[i + j]);
                if (j != n - 1) sb.append(" ");
            }
            String ngram = sb.toString();
            map.put(ngram, map.getOrDefault(ngram, 0) + 1);
        }
        return map;
    }

    // En sık geçen ilk 10 n-gram'ı yazdır
    public static void printTop10(Map<String, Integer> map) {
        map.entrySet()
           .stream()
           .sorted((e1, e2) -> e2.getValue().compareTo(e1.getValue()))
           .limit(10)
           .forEach(entry -> System.out.println(entry.getKey() + " : " + entry.getValue()));
    }
}
```

NGramGenerator Class

```java
package nlp1;

import java.io.BufferedReader;
import java.io.FileReader;

public class FileProcessor {

    // Dosyayı oku ve tek string olarak döndür
    public static String readFile(String filePath) throws Exception {
        BufferedReader reader = new BufferedReader(new FileReader(filePath));
        StringBuilder sb = new StringBuilder();
        String line;
        while ((line = reader.readLine()) != null) {
            sb.append(line).append(" ");
        }
        reader.close();
        return sb.toString();
    }

    // Metni temizle (noktalama çıkar, küçük harfe çevir, boşlukla böl)
    public static String[] preprocess(String text) {
        text = text.toLowerCase().replaceAll("[^a-zA-ZçğıöşüÇĞİÖŞÜ\\s]", "");
        return text.trim().split("\\s+");
    }
}
```

FileProcessor Class

Output for grimms-fairy-tales_1

```
=== 1-gram Analysis ===

the : 227
and : 163
to : 91
he : 74
you : 43
was : 39
his : 38
it : 35
a : 34
him : 34
Running time: 10 ms


=== 2-gram Analysis ===

the fox : 22
to the : 22
and the : 19
the golden : 17
in the : 17
the king : 13
the bird : 13
the princess : 12
then the : 11
but the : 11
Running time: 4 ms
```

```
=== 3-gram Analysis ===

and in the : 7
the golden bird : 7
came to the : 7
the young man : 7
to the king : 6
you come to : 5
the king and : 5
in the morning : 5
the fox said : 5
then the fox : 5
Running time: 5 ms
```

Test Results

grimms-fairy-tales_1

=== 1-gram Analysis ===

the : 227
and : 163
to : 91
he : 74
you : 43
was : 39
his : 38
it : 35
a : 34
him : 34

Running time: 10 ms

=== 2-gram Analysis ===

the fox : 22
to the : 22
and the : 19
the golden : 17
in the : 17
the king : 13
the bird : 13
the princess : 12
then the : 11
but the : 11

Running time: 4 ms

=== 3-gram Analysis ===

and in the : 7
the golden bird : 7
came to the : 7
the young man : 7
to the king : 6
you come to : 5
the king and : 5
in the morning : 5
the fox said : 5
then the fox : 5

Running time: 5 ms

grimms-fairy-tales_2

=== 1-gram Analysis ===

the : 96
and : 86
a : 74
he : 60
ı : 55
to : 52
you : 49
his : 48
it : 42
for : 34

Running time: 9 ms

=== 2-gram Analysis ===

as he : 10
said the : 9
it is : 7
he was : 7
said hans : 7
a pig : 6
the cow : 6
he had : 6
to be : 6
the world : 6

Running time: 3 ms

=== 3-gram Analysis ===

ı gave a : 4
for it and : 4
butter and cheese : 3
it and the : 3
gave him a : 3
by his side : 2
and at last : 2
seven long years : 2
said the other : 2
ı will change : 2

Running time: 4 ms

grimms-fairy-tales_3

=== 1-gram Analysis ===

the : 71
and : 54
he : 47
a : 29
to : 23
in : 22
was : 19
her : 18
of : 17
that : 16

Running time: 8 ms

=== 2-gram Analysis ===

in the : 12
flower and : 7
the flower : 6
could not : 6
the castle : 6

to the : 5
then he : 5
the fairy : 5
he could : 4
into a : 4

Running time: 1 ms

=== 3-gram Analysis ===

the flower and : 4
the middle of : 3
with the flower : 3
could not move : 3
he could not : 3
in the middle : 3
in the castle : 3
whose name was : 2
whu tu whu : 2
changed into a : 2

Running time: 1 ms

grimms-fairy-tales_4

=== 1-gram Analysis ===

the : 115
and : 60

a : 40
to : 38
they : 26
in : 19
had : 19
with : 18
of : 18
ass : 16

Running time: 7 ms

=== 2-gram Analysis ===

the ass : 15
said the : 13
the cock : 8
the cat : 8
the dog : 8
in the : 8
to the : 7
and the : 7
of the : 6
as they : 5

Running time: 2 ms

=== 3-gram Analysis ===

said the ass : 8
the great city : 4
and the cock : 4
up to the : 3
said the cock : 3

top of the : 3
the house and : 3
him in the : 2
all his might : 2
because ı am : 2

Running time: 2 ms

grimms-fairy-tales_5

=== 1-gram Analysis ===

the : 59
and : 53
to : 25
his : 21
a : 17
he : 17
him : 15
sultan : 15
wolf : 13
was : 12

Running time: 8 ms

=== 2-gram Analysis ===

the wolf : 13
in the : 7
the shepherd : 6
and the : 5
his wife : 5
ı will : 5
old sultan : 4
the boar : 4
to be : 4
and when : 3

Running time: 2 ms

=== 3-gram Analysis ===

the shepherd and : 3
shepherd and his : 3
and his wife : 3
the wolf was : 2
as long as : 2
he has served : 2
in the air : 2
sultan who was : 2
out of the : 2
has served us : 2

Running time: 1 ms

grimms-fairy-tales_6

=== 1-gram Analysis ===

the : 38
and : 25
a : 17
to : 13
straw : 8
she : 8
her : 8
have : 8
on : 8
had : 7

Running time: 6 ms

=== 2-gram Analysis ===

the straw : 6
the bean : 5
the coal : 5
into the : 4
have been : 4
to the : 3
when she : 3
old woman : 3
down to : 2
would have : 2

Running time: 1 ms

=== 3-gram Analysis ===

would have been : 2
the straw the : 2
should have been : 2
the old woman : 2
into the pan : 2
idea and said : 1
her last the : 1
good fortune a : 1
made a fire : 1
then have a : 1

Running time: 1 ms

grimms-fairy-tales_7

=== 1-gram Analysis ===

the : 141
and : 105
a : 33
to : 29
was : 27
in : 25

her : 25
that : 22
of : 21
on : 21

Running time: 9 ms

=== 2-gram Analysis ===

and the : 14
in the : 12
on the : 10
of the : 10
king and : 7
the king : 7
and queen : 6
all the : 6
into the : 5
to the : 5

Running time: 3 ms

=== 3-gram Analysis ===

king and queen : 6
the king and : 5
and the cook : 3
and the horses : 3
till at last : 3
the kings daughter : 3
came to the : 2

and queen had : 2
the palace and : 2
that he could : 2

Running time: 4 ms

BİLİM İŞ BAŞINDA

=== 1-gram Analysis ===

bir : 454
ve : 202
bu : 154
için : 90
de : 77
çok : 73
ile : 71
daha : 66
da : 60
olarak : 52

Running time: 18 ms

=== 2-gram Analysis ===

bir biçimde : 19
pek çok : 19
ile ilgili : 11
deniz suyu : 10
bir elektrik : 9

herhangi bir : 9
uygun bir : 9
yine de : 9
iyi bir : 9
bu nedenle : 9

Running time: 10 ms

=== 3-gram Analysis ===

yaklaşık yıl önce : 6
ne yazık ki : 5
başka herhangi bir : 4
söz konusu olduğunda : 4
diğer bir deyişle : 4
dahil olmak üzere : 4
bir kabın içine : 3
bir elektrik yükü : 3
buzun erime noktası : 3
yaygın olarak kullanılan : 3

Running time: 11 ms

DEĞİŞİM

=== 1-gram Analysis ===

bir : 541
ve : 256
bu : 164
gregor : 159
ama : 146
de : 123
da : 103
gregorun : 95
gibi : 95
için : 93

Running time: 23 ms

=== 2-gram Analysis ===

bir şey : 29
anne ve : 29
müdür bey : 24
bay samsa : 18
müdür beyin : 18
bundan böyle : 17
hiç de : 15
ya da : 15
bir an : 15
böyle bir : 14

Running time: 18 ms

=== 3-gram Analysis ===

baylardan orta boylusu : 9
anne ve babası : 9
ne var ki : 8
en ufak bir : 6
diye geçirdi içinden : 5
anne ve babasına : 5
ama yine de : 4
böyle bir şeyin : 4
anne ve babasının : 4
üzerine yığılıp kaldı : 3

Running time: 12 ms